

PART 3

Setup & Folder Structure

FinVault – MERN Stack Virtual Stock Trading Platform

DOCUMENT INFORMATION

Project Title	FinVault – Virtual Stock Trading Platform
Document Phase	Phase 1: Project Documentation
Part	Part 3 – Setup & Folder Structure
Author	Ganesh Basode
Affiliation	Department of Computer Science & Engineering
Institution	G. Pullaiah College of Engineering and Technology (GPCET)
Date	July 2026
Status	Final Version

1. Workspace Directory Layout

FinVault utilizes a split directory design where the backend and frontend run in parallel configurations.

```
FinVault/
├── package.json
├── package-lock.json
├── backend/
│   ├── config/           # DB connection config
│   ├── controllers/      # Endpoint business controllers
│   ├── middleware/       # Security filters & upload config
│   ├── models/           # Mongoose DB models
│   ├── routes/           # Express routing definitions
│   ├── scripts/          # Internal helper scripts
│   ├── seed/             # Seed database configuration files
│   ├── uploads/          # Destination folder for uploaded images
│   ├── package.json      # Backend node dependencies
│   ├── seedStocks.js     # Executable stock database loader
│   └── server.js         # Express server initialization entry
├── frontend/
│   ├── src/
│   │   ├── admin/       # Admin panel view modules
│   │   ├── api/         # API client setup modules
│   │   ├── assets/       # Static assets & styling images
│   │   ├── components/   # Reusable UI React widgets
│   │   ├── context/      # AuthContext provider
│   │   ├── css/          # Styles sheets
│   │   ├── pages/        # Base pages of SPA
│   │   ├── App.jsx       # Route layout component
│   │   └── main.jsx      # Application entry
│   ├── package.json      # Frontend node dependencies
│   └── vite.config.js    # Vite build configurations
```

2. Prerequisites & Local Environment

Before starting, ensure you have the following system software installed:

- **Node.js:** Version 18.x or above.
- **npm:** Node Package Manager v9.x or above.
- **MongoDB:** A running local MongoDB community server, or a valid connection URI to a MongoDB Atlas cluster.

3. Backend Setup Guide

3.1 Installation

Open a terminal, move into the backend folder, and install the npm dependencies:

```
cd backend
npm install
```

3.2 Configuration (.env)

Create a new text file named `.env` inside the backend directory. Fill in your environment parameters as shown below:

```
PORT=5001
MONGO_URI=mongodb://127.0.0.1:27017/SBStocks
JWT_SECRET=FinVault@123
```

3.3 Database Seeding

Run the stock seeding script to establish initial mock stock data (such as AAPL, GOOG, TSLA) in your MongoDB collection:

```
node seedStocks.js
```

3.4 Starting the Server

Launch the Express API server:

```
node server.js
```

On success, the console will print: ■ **Server running on http://localhost:5001** and confirm database connection.

4. Frontend Setup Guide

4.1 Installation & Run

Open another terminal tab, navigate into the frontend directory, install dependencies, and run the development server:

```
cd frontend
npm install
npm run dev
```

4.2 System Access

Vite will run the React client. Open your browser and navigate to: **http://localhost:5173**.

All requests are automatically routed to the backend port (`http://localhost:5001`) as configured in the api client layers.